

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			137 (1) 55 (1)		2		2		
	(b)	(i)		Neutron don't accept the symbol	1			1		
		(ii)		Only <u>one</u> neutron causes further fission or the other <u>two</u> can't produce further fission (1) To prevent uncontrolled <u>chain</u> reaction or to prevent overheating or to prevent meltdown / to control the <u>chain</u> reaction (1) or converse argument. Don't accept to stop the chain reaction or to stop an explosion	2			2		
	(c)			Any 2 x(1): - It is radioactive for a long time or products have long half-lives - It is ionising - It is penetrating	2			2		
	(d)	(i)		It is 0.3 % / it is a very small % / one of the smallest parts Don't accept it's the smallest		1		1		
		(ii)		Nuclear decay is a random process (1) Take measurements over a long time [so second group is better] or the <u>second group's results</u> are more accurate or the <u>second group</u> can calculate a mean (1)	2			2		2
				Question 5 total	7	3	0	10	0	2